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Amazon Stock Forecasting Rubric

Why am I doing this?

This case study allows you to apply time series analysis and machine learning techniques to a real-world financial dataset. You will explore Amazon stock price data and evaluate how different modeling approaches perform when forecasting complex, non-linear time series data. Through this project, you will gain experience working with financial data, building predictive models, and interpreting results in a real-world context where uncertainty and volatility are important factors.

What am I going to do?

The GitHub repository for this case study contains Amazon (AMZN) stock data from 1997–2026, along with starter code and supporting materials.

You will:

- Perform exploratory data analysis (EDA) on stock price data
- Prepare the data for time series modeling (including transformations)
- Build and evaluate an ARIMA model
- Build and evaluate an LSTM model
- Compare model performance using evaluation metrics
- Interpret results and provide a final recommendation

The dataset includes daily observations such as closing price and trading volume, and captures major market events like the dot-com crash, 2008 financial crisis, and COVID-19 volatility.

Your final deliverables should include:

- A 2–3 page report (PDF)
- A GitHub repository containing:
 - Code (Jupyter Notebook)
 - Dataset
 - Outputs/visualizations
 - README file
 - References

Tips for success:

- Carefully examine time series plots to identify trends and structural changes
- Use transformations (e.g., log scaling) when appropriate to stabilize variance
- Ensure models are applied correctly (e.g., stationarity for ARIMA)

- Compare models using multiple evaluation metrics, not just one
- Focus on interpretation, not just producing results

How will I know I have succeeded?

You will meet expectations when you successfully complete the criteria below:

Spec Category	Spec Details
Formatting	• One GitHub repository (submitted via link on Canvas) • Must include: ◦ A README.md file (which auto displays) ◦ A LICENSE.md file (use MIT as default) ◦ A SCRIPTS folder ◦ A DATA folder ◦ An OUTPUT folder
README.md	Goal: This file serves as an orientation to everyone who comes to your repository. It should enable them to get their bearings. Use markdown headers to divide content. Section 1: Software and Platform • Software used (Python, Colab, Jupyter Notebook) • Required packages (pandas, numpy, matplotlib, statsmodels, torch, scikit-learn, etc.) • Platform used (Mac, Windows, or Linux) Section 2: Map of Documentation • Provide a folder structure outline (DATA, SCRIPTS, OUTPUT) • List all files within each folder Section 3: Instructions for Reproducing Results • Provide step-by-step instructions in plain English • Include downloading data, installing

	packages, running <code>official_amazon.ipynb</code>, and generating outputs
LICENSE.md	Goal: This file explains to a visitor the terms under which they may use and cite your repository. • Select a license from GitHub options • MIT license is recommended
SCRIPTS folder	Goal: This folder contains all the source code for your project. • Include all scripts used (e.g. <code>official_amazon.ipynb</code>) • Name scripts clearly and logically • Include header comments describing purpose, inputs, and outputs • Include detailed comments explaining preprocessing, ARIMA model, LSTM model, and evaluation steps
DATA folder	Goal: This folder contains all of the data for this project. • Include dataset (<code>amazon_data.csv</code>) • Include Data Appendix PDF The Data Appendix should include: • Unit of observation (each row = one trading day) • Variable descriptions (Date, Open, High, Low, Close, Volume) • Transformations (log variables, scaling) If data is too large, include instructions to obtain it
OUTPUT folder	Goal: This folder contains all output generated by the project. • Include figures, tables, and results • Examples: ○ Time series plots

	○ Distribution plots ○ ACF/PACF plots ○ Forecast vs actual plots ● Use clear and informative file names
References	All references should be listed at the end of the document. ● Use IEEE documentation style ● Include sources for ARIMA, LSTM, and dataset ● Provide complete citations
Report	Goal: The final report should clearly communicate the analysis, modeling process, results, and final recommendation. ● Submit a 2–3 page PDF report ● Include a summary of the dataset and forecasting objective ● Discuss exploratory data analysis and preprocessing steps ● Explain ARIMA and LSTM model implementation ● Compare model performance using evaluation metrics (MAE, RMSE, MAPE) ● Interpret results clearly and provide a final recommendation on which model performed better and why ● Include figures/tables from the OUTPUT folder when appropriate ● Writing should be organized, concise, and professional